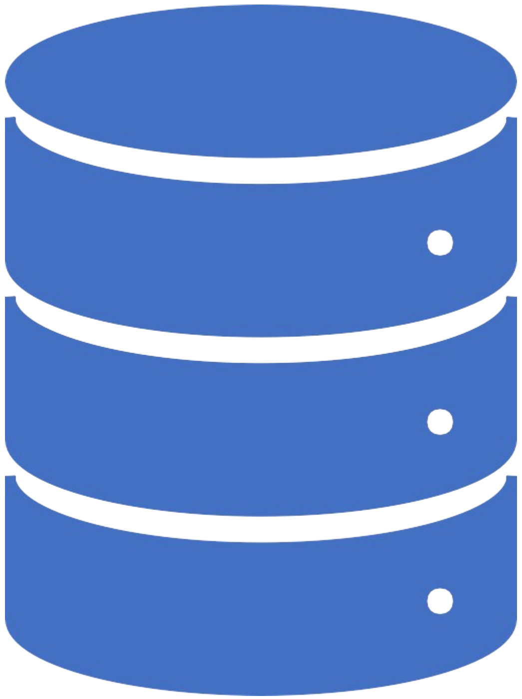


Database Systems
[*Introduction to Database*]

What is Data?

- **Data** is a collection of a distinct small unit of information.
  - It can be used in a variety of forms like text,
    numbers, media, bytes, etc.
  - It can be stored in pieces of paper or electronic memory, etc.
- Word '***Data***' is originated from the word '***datum***' that means '***single piece of information***.'
- It is plural of the word datum.
- In computing, Data is information that can be translated into a form for efficient movement and processing.
- Data is interchangeable.

What is Database?

- A **database** is an organized collection of data, so that it can be easily accessed and managed.
- You can organize data into tables, rows, columns, and index it to make it easier to find relevant information.
- **Database handlers** create a database in such a way that only one set of software program provides access of data to all the users.
- The **main purpose** of the database is to operate a large amount of information by storing, retrieving, and managing data.
- There are many **dynamic websites** on the World Wide Web nowadays which are handled through databases.
  - For example, a model that checks the availability of rooms in a hotel.
  - It is an example of a dynamic website that uses a database.

What is Database?

- There are many **databases available** like MySQL, Sybase, Oracle, MongoDB, Informix, PostgreSQL, SQL Server, etc.
- Modern databases are managed by the DataBase Management System (DBMS).
- **SQL** (Structured Query Language) is used to operate
on the data stored in a database.
- SQL depends on relational algebra and tuple relational calculus.

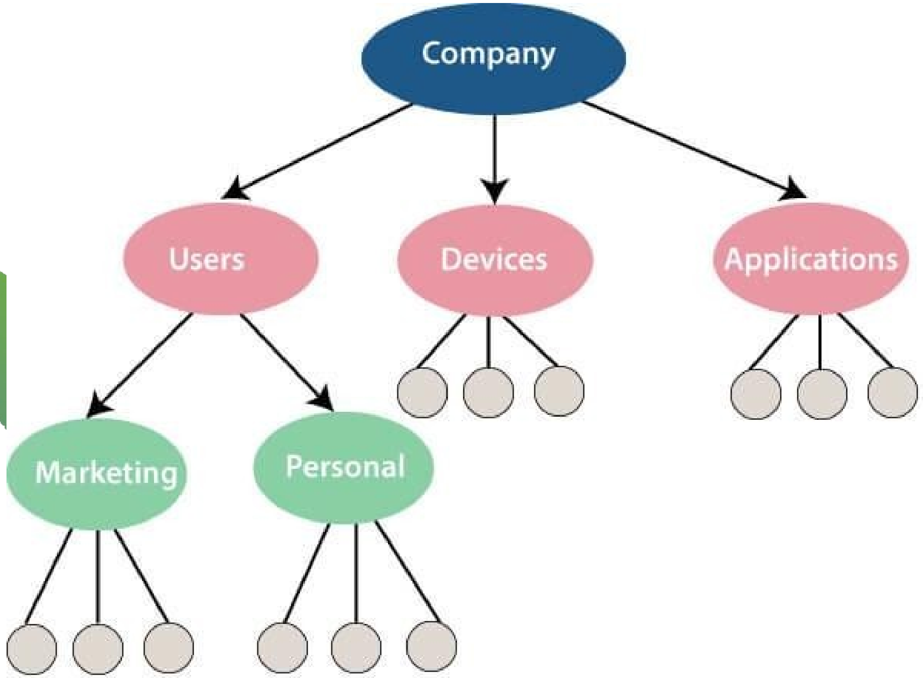
Evolution of Databases

- The database has completed more than 50 years of journey of its evolution from flat-file system to relational and objects relational systems.
- It has gone through several generations.

The Evolution : File-Based

- 1968 was the year when File-Based database were introduced.
- In file-based databases, data was maintained in a flat file. Though files have many advantages, there are several limitations.
- One of the major advantages is that the file system has various access methods, e.g., sequential, indexed, and random.
- It requires extensive programming in a third-generation language such as COBOL, BASIC.

The Evolution : Hierarchical Data Model (1968-



- Prominent hierarchical database model was IBM's first DBMS.
- It was called IMS (Information Management System).
- In this model, files are related in a parent/child manner.
- Small circle represents objects.
- Like file system, this model also had some limitations like complex implementation, lack structural independence, can't easily handle a many-many relationship, etc.

The Evolution : Network data model

- **Charles Bachman** developed the first DBMS at Honeywell called Integrated Data Store (IDS).
- It was developed in the early 1960s, but it was standardized in 1971 by the CODASYL group (Conference on Data Systems Languages).
- In this model, files are related as owners and members, like to the common network model.
- Network data model identified the following components:
    - Network schema (Database organization)
    - Sub-schema (views of database per user)
    - Data management language (procedural)
- This model also had some limitations like system complexity and difficult to design and maintain.

The Evolution : Relational Database (1970 – Present)

- It is the era of Relational Database and Database Management.
- In 1970, the relational model was proposed by E.F. Codd.
- Relational database model has two main terminologies called ***instance*** and ***schema***.
  - The instance is a table with rows or columns
  - Schema specifies the structure like name of the relation, type of each column and name.
- This model uses some mathematical concept like set theory and predicate logic.
- The first internet database application had been created in 1995.
- During the era of the relational database, many more models had introduced like object-oriented model, object-relational model, etc.

The Evolution : Cloud database

- Cloud database facilitates you to store, manage, and retrieve their structured, unstructured data via a cloud platform.
- This data is accessible over the Internet.
- Cloud databases are also called a database as service (DBaaS) because they are offered as a managed service.
- Some best cloud options are:
    - AWS (Amazon Web Services)
    - Snowflake Computing
    - Oracle Database Cloud Services
    - Microsoft SQL server
    - Google cloud spanner

The Evolution : Cloud database (Advantages)

- **Lower costs:** Generally, company provider does not have to invest in databases. It can maintain and support one or more data centers.
- **Automated:** Cloud databases are enriched with a variety of automated processes such as recovery, failover, and auto-scaling.
- **Increased Accessibility:** You can access your cloud-based database from any location, anytime. All you need is just an internet connection.

The Evolution : NoSQL Database

- A NoSQL (not only SQL) database is an approach to design such databases that can accommodate a wide variety of data models.
- It is an alternative to traditional relational databases in which data is placed in tables, and data schema is perfectly designed before the database is built.
- NoSQL databases are useful for a large set of distributed data.
- Some examples of NoSQL database system with their category are:
  - MongoDB, CouchDB, Cloudant **(Document-based)**
  - Memcached, Redis, Coherence **(key-value store)**
  - HBase, Big Table, Accumulo **(Tabular)**

The Evolution : NoSQL Database (Advantage)

High Scalability:

- NoSQL can handle an extensive amount of data because of scalability. If the data grows, NoSQL database scale it to handle that data in an efficient manner.

High Availability:

- NoSQL supports auto replication. Auto replication makes it highly available because, in case of any failure, data replicates itself to the previous consistent state.

The Evolution : NoSQL Database (Disadvantage)

- **Open source:**
  - NoSQL is an open-source database, so there is no reliable standard for NoSQL yet.
- **Management challenge:**
  - Data management in NoSQL is much more complicated than relational databases. It is very challenging to install and even more hectic to manage daily.
- **GUI is not available:**
  - GUI tools for NoSQL database are not easily available in the market.
- **Backup:**
  - Backup is a great weak point for NoSQL databases. Some databases, like MongoDB, have no powerful approaches for data backup.

The Evolution : The Object-Oriented Databases

- The object-oriented databases contain data in the form of object and classes.
- Objects are the real-world entity, and types are the collection
  of objects.
- An object-oriented database is a combination of relational model features with objects-oriented principles.
- It is an alternative implementation to that of the relational model.
- Object-oriented databases hold the rules of obje
  programming.
- An object-oriented database manageme   d
  hybrid
  application.

The Evolution : The Object-Oriented Databases (Properties)

Object-oriented programming

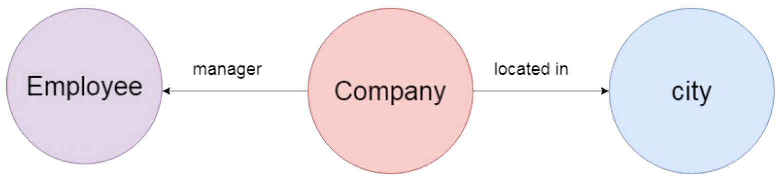
- Objects
- Classes
- Inheritance
- Polymorphism
- Encapsulation

Relational database

- Atomicity
- Consistency
- Integrity
- Durability
- Concurrency
- Query processing

The Evolution : Graph Databases

- A graph database is a NoSQL database. It is a graphical representation of data. It contains nodes and edges. A node represents an entity, and each edge represents a relationship between two edges. Every node in a graph database represents a unique identifier.
- Graph databases are beneficial for searching the relationship between data because they highlight the relationship between relevant data.
- Graph databases are very useful when the database contains a complex relationship and dynamic schema.
- It is mostly used in **supply chain management**, identifying the source of IP telephony.



The Evolution : DBMS

- DataBase Management System is software which is used to store and retrieve the database.
- For example, Oracle, MySQL, etc.; these are some popular DBMS tools.
  - DBMS provides the interface to perform the various operations like creation, deletion, modification, etc.
  - DBMS allows the user to create their databases as per their requirement.
  - DBMS accepts the request from the application and provides specific data through the operating system.
  - DBMS contains the group of programs which acts according to the user instruction.
  - It provides security to the database.

Advantage of DBMS

Controls redundancy:

- It stores all the data in a single database file, so it can control data redundancy.

Data sharing:

- An authorized user can share the data among multiple users.

Backup:

- It provides Backup and recovery subsystem. This recovery system creates automatic data from system failure and restores data if required.

Multiple user interfaces:

- It provides a different type of user interfaces like GUI, application interfaces.

Disadvantage of DBMS

Size:

- It occupies large disk space and large memory to run efficiently.

Cost:

- DBMS requires a high-speed data processor and larger memory to run DBMS software, so it is costly.

Complexity:

- DBMS creates additional complexity and requirements.

The Evolution : RDBMS

- The word RDBMS is termed as 'Relational Database Management System.' It is represented as a table that contains rows and column.
- RDBMS is based on the Relational model; it was introduced by E. F. Codd.
- A relational database contains the following components:
  - Table
  - Record/ Tuple
  - Field/Column name /Attribute
  - Instance
  - Schema
  - Keys
- An RDBMS is a tabular DBMS that maintains the secu accuracy, and consistency of the data.